Visual Analytics Project Proposal

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General idea

We are proposing to develop a visualization system with the purpose of providing general insights on the topic of vehicle accidents in Italy.

Intended users

The intended users for this system would be members of the Ministry of Transport, helping them with taking informed decision on how to better prevent accidents in Italy.

Dataset

The dataset in use contains details about vehicle accidents occurred in Italy during 2024. This dataset has been dowloaded from the ISTAT website. The full dataset comprises about 500.000 tuples for 14 attributes and it can be reduced by removing some included aggregations that can be recomputed if necessary.

The attributes in the dataset are in summary: location (province) and time (month) of accident, road type and section of accident (motorway, traffic circle, etc.), type of accident (between vehicles, vehicle-pedestrian, etc), type of vehicle involved (car, motorcycle, etc.), and number of accidents registered.

More details are provided with a table at the end of the proposal.

Visualization System Description

The system is composed of two main sections as shown in Figure 1.

In section A the number of accidents for each main attribute in the dataset it's visualized. This is is done with a number of bar charts that use the categorical values on the bottom axis and show the number of accidents on the vertical axis. These charts are coordinated beetween each other. As an example, selecting the accidents that occurred between January and March also highlights the road type and the kinds of vehicles involved.

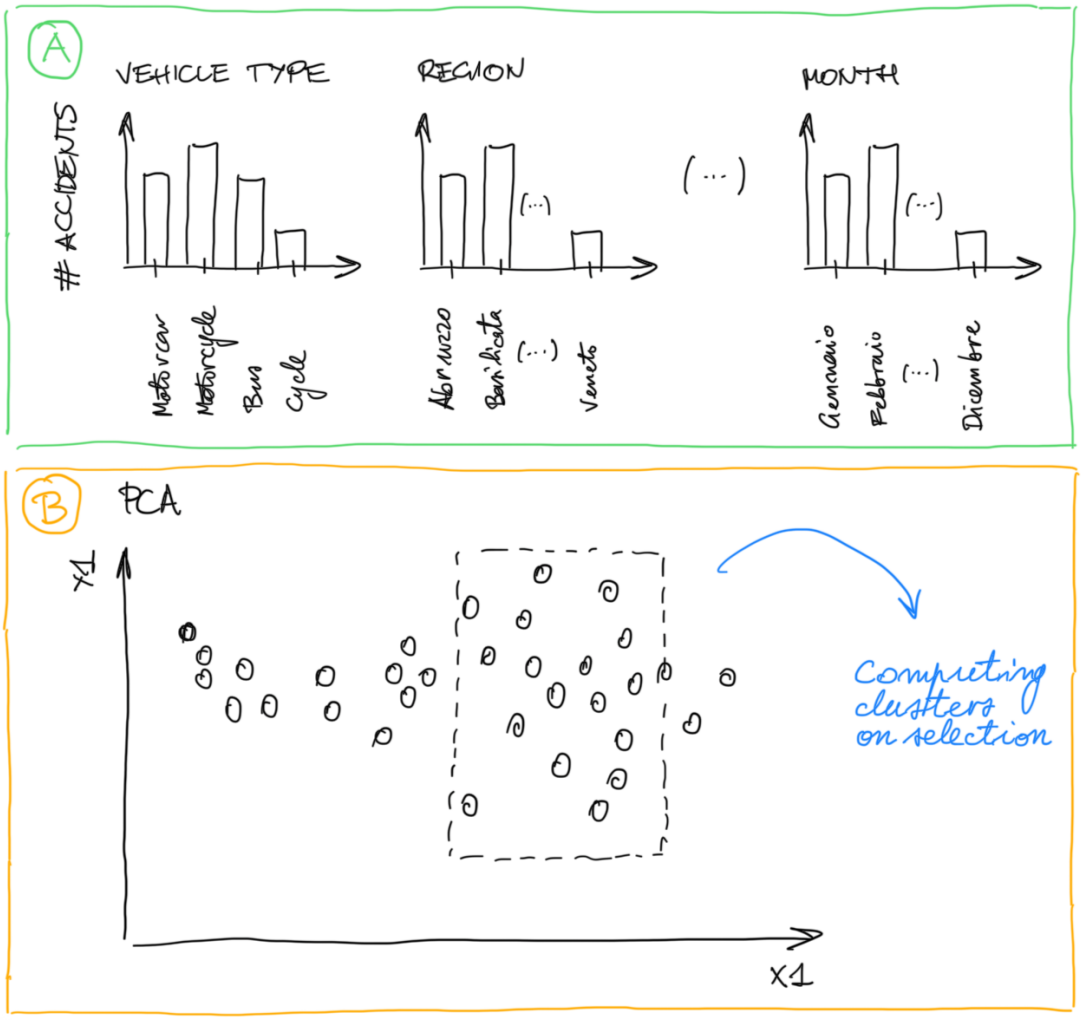

Figure 1: Draft of the visualization system

In section B, the result of PCA over the dataset is presented to the user. Data points can be colored according to the categorical attributes of the dataset: vehicle type, road type etc. The PCA graph is coordinated with the graphs in section A. As an example, selecting the accidents between January and March on section A higlights the relative data points on the PCA graph, and vice versa. Furthermore, the user can visually trigger an analytic by requesting the system to find clusters (*e.g. k-means*) in a selection over the PCA graph.

Dataset Details

Attribute name	Description
Area code	ID of Region or Province
Area description	Name of Region or Province
Road type code	Number identifying a road type. Identifies motorways, urban roads and other type
Road type description	Textual version of the attribute above
Road section code	Number identifying a road section. Identifies: tunnel, bend, crossroad, traffic circle, bump/slope or bottleneck, straight stretch, level crossing.
Road section description	Textual version of the attribute above
Accident type code	Number identifying the accident type. Identifies: accidents between vehicles, vehicle-pedestrian accients and accidents involving a single vehicle.
Accident type descripiton	Textual version of the attribute above
Vehicle type code	Number identifying the vehicle involved in the accident. Identifies: truks and motor units, cycles, tram, other vehicles, bus or trolley-bus, mini cars, three-wheeler or motor-van, motorcars, motorcycles, mopeds
Vehicle type description	Textual version of the attribute above
Month number	Month number
Month name	Month name
Year	Year number. Dataset contains only data for 2024
Observations	Number of accidents occurred with the specified attributes